# APACHE HADOOP MAPREDUCE

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Semester: 7th

## Aim:

To write a MapReduce program in Python using the mrjob library that reads student marks stored in HDFS and calculates the average marks, maximum marks, and minimum marks for each subject.

## Problem Statement:

An educational institution stores student marks in a CSV file containing the student ID, student name, subject, and marks. The dataset is stored in HDFS for distributed processing. The objective is to write a MapReduce program using Python's mrjob framework that reads the student marks, groups the records by subject, and calculates the average, maximum, and minimum marks for every subject.

In the Map phase, each record is read and the subject and marks are extracted. The Mapper outputs the pair (subject, marks). Hadoop then groups all marks belonging to the same subject. In the Reduce phase, the grouped marks are used to calculate the total, count, average, maximum, and minimum.

## Tools and Environment Used:

Operating System: Ubuntu 24.04 LTS (virtual machine)

Java: OpenJDK 17

Hadoop: Apache Hadoop 3.5.0

Language: Python 3.12

Library: mrjob

## Code Implementation:

File name: education_mapreduce.py

```python
from mrjob.job import MRJob


class StudentPerformance(MRJob):

    def mapper(self, _, line):
        student_id, name, subject, marks = line.split(",")

        yield subject, int(marks)

    def reducer(self, subject, marks):
        marks = list(marks)

        average = sum(marks) / len(marks)
        maximum = max(marks)
        minimum = min(marks)

        yield subject, {
            "average": round(average, 2),
            "maximum": maximum,
            "minimum": minimum
        }


if __name__ == '__main__':
    StudentPerformance.run()
```

## Mapper and Reducer Explanation:

mapper(): Reads each CSV line, extracts the subject and marks, converts the marks to a numeric value, and outputs (subject, marks).

reducer(): Receives one subject with all its marks and calculates the average, maximum, and minimum marks.

run(): Starts the MapReduce job. The option -r hadoop makes mrjob submit the job to the Hadoop cluster through Hadoop Streaming.

## Input Data:

The student marks file (student_marks.csv) contains the following records:

```
aiml@aiml-HP-Pro:~/education_mapreduce$ hdfs dfs -cat /user/$USER/education/student_marks.csv
S001,Rahul,Math,80
S002,Anita,Math,90
S003,Ravi,Math,70
S004,Priya,Math,85
S001,Rahul,Science,85
S002,Anita,Science,95
S003,Ravi,Science,75
S004,Priya,Science,85
S001,Rahul,English,70
S002,Anita,English,80
S003,Ravi,English,90
S004,Priya,English,80
```

The file was stored in HDFS at: /user/aiml/education/student_marks.csv

## Execution Steps:

1. Start the Hadoop services and check that they are running:

```
start-dfs.sh
start-yarn.sh
jps
```

2. Activate the Python environment that has mrjob installed:

```
source ~/mrjob-env/bin/activate
```

3. Create a directory in HDFS and upload the student marks file:

```
hdfs dfs -mkdir -p /user/aiml/education
hdfs dfs -put -f ~/student_marks.csv /user/aiml/education/
hdfs dfs -ls /user/aiml/education/
```

4. Verify the input file stored in HDFS:

```
hdfs dfs -cat /user/aiml/education/student_marks.csv
```

5. Run the MapReduce program on Hadoop:

```
python ~/education_mapreduce.py -r hadoop hdfs:///user/aiml/education/student_marks.csv
```

## Execution Output:

**Checking that all Hadoop services are running (jps):**

```
aiml@aiml-HP-Pro:~/education_mapreduce$ jps
8082 DataNode
7877 NameNode
26422 Jps
8315 SecondaryNameNode
8571 ResourceManager
8926 NodeManager
```

Figure 1: Hadoop services running - NameNode, DataNode, SecondaryNameNode, ResourceManager, NodeManager

**Uploading the student marks file to HDFS and viewing it:**

```
aiml@aiml-HP-Pro:~/education_mapreduce$ hdfs dfs -ls /user/$USER/education
Found 4 items
drwxr-xr-x   - aiml supergroup          0 2026-09-25 12:15 /user/aiml/education/output
drwxr-xr-x   - aiml supergroup          0 2026-09-25 12:21 /user/aiml/education/result
drwxr-xr-x   - aiml supergroup          0 2026-09-25 12:47 /user/aiml/education/result.txt
-rw-r--r--   3 aiml supergroup        249 2026-09-25 12:10 /user/aiml/education/student_marks.csv
```

Figure 2: student_marks.csv stored in HDFS at /user/aiml/education/

**Running the MapReduce job on Hadoop (YARN):**

```
aiml@aiml-HP-Pro:~/education_mapreduce$ python3 education.py \
-r hadoop \
hdfs:///user/$USER/education/student_marks.csv
No configs found; falling back on auto-configuration
No configs specified for hadoop runner
Looking for hadoop binary in /usr/local/hadoop/bin...
Found hadoop binary: /usr/local/hadoop/bin/hadoop
Using Hadoop version 3.4.2
Looking for Hadoop streaming jar in /usr/local/hadoop...
Found Hadoop streaming jar: /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.4.2.jar
Creating temp directory /tmp/education.aiml.20260925.065032.477108
uploading working dir files to hdfs:///user/aiml/tmp/mrjob/education.aiml.20260925.065032.477108/files/wd...
Copying other local files to hdfs:///user/aiml/tmp/mrjob/education.aiml.20260925.065032.477108/files/
Running step 1 of 1...
  packageJobJar: [/tmp/hadoop-unjar11325694520773439356/] [] /tmp/streamjob12584245839343151240.jar tmpDir=null
  Connecting to ResourceManager at /0.0.0.0:8032
  Connecting to ResourceManager at /0.0.0.0:8032
  Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/aiml/.staging/job_1790314469432_0005
  Total input files to process : 1
  number of splits:2
  Submitting tokens for job: job_1790314469432_0005
  Executing with tokens: []
  resource-types.xml not found
  Unable to find 'resource-types.xml'.
  Submitted application application_1790314469432_0005
  The url to track the job: http://aiml-HP-Pro:8088/proxy/application_1790314469432_0005/
  Running job: job_1790314469432_0005
  Job job_1790314469432_0005 running in uber mode : false
   map 0% reduce 0%
   map 100% reduce 0%
   map 100% reduce 100%
  Job job_1790314469432_0005 completed successfully
  Output directory: hdfs:///user/aiml/tmp/mrjob/education.aiml.20260925.065032.477108/output
Counters: 54
```

Figure 3: MapReduce job completed successfully.

## Job Output:

The reducer groups all marks for each subject and calculates the required statistics.

```
aiml@aiml-HP-Pro:~/education_mapreduce$ hdfs dfs -cat /user/$USER/education/result.txt/part-00000
"English"       {"average": 80.0, "maximum": 90, "minimum": 70}
"Math"  {"average": 81.25, "maximum": 90, "minimum": 70}
"Science"       {"average": 85.0, "maximum": 95, "minimum": 75}
```

## Brief Discussion:

The program works in two stages. In the Map stage, every student record is read and the subject and marks are extracted. Hadoop then performs the Shuffle and Sort operation and groups all marks having the same subject.

In the Reduce stage, all marks belonging to one subject are collected. The reducer calculates the total and count, then uses them to obtain the average. It also identifies the maximum and minimum marks.

For Math: 80 + 90 + 70 + 85 = 325; Average = 325 / 4 = 81.25; Maximum = 90; Minimum = 70.

For Science: 85 + 95 + 75 + 85 = 340; Average = 340 / 4 = 85.00; Maximum = 95; Minimum = 75.

For English: 70 + 80 + 90 + 80 = 320; Average = 320 / 4 = 80.00; Maximum = 90; Minimum = 70.

## Result:

The Hadoop MapReduce job successfully calculated the average, maximum, and minimum marks for each subject.

| Subject | Average | Maximum | Minimum |
|---|---|---|---|
| English | 80.0 | 90 | 70 |
| Math | 81.25 | 90 | 70 |
| Science | 85.0 | 95 | 75 |

## Conclusion:

The Hadoop MapReduce program was successfully implemented using Python and mrjob on a single-node Hadoop cluster running on Ubuntu. The student marks dataset was stored in HDFS and processed using MapReduce on YARN. The program correctly calculated the average, maximum, and minimum marks for each subject: English, Math, and Science.